

05_Python EDA

Objective

The objective of the Python Exploratory Data Analysis (EDA) phase was to develop an initial understanding of marketplace performance using the cleaned and feature-engineered datasets.

The analysis focused on identifying high-level patterns related to customer activity, order performance, logistics efficiency, customer satisfaction, revenue growth, and geographic distribution.

The insights generated during this phase established a foundation for deeper business analysis in PostgreSQL.

Analysis Areas

The following business areas were explored:

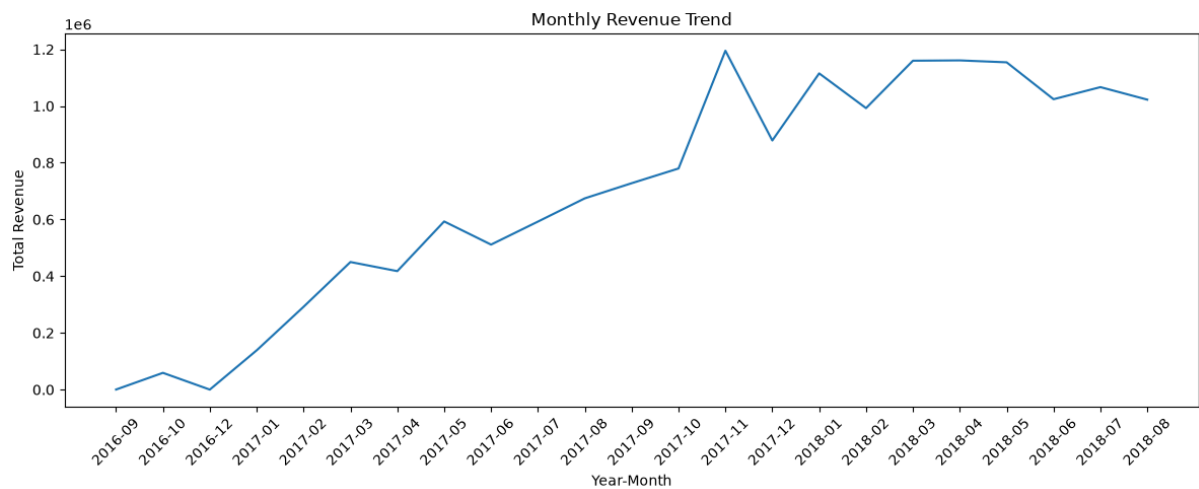
- Order Trends
- Order Status Distribution
- Delivery Performance
- Customer Reviews
- Revenue Trends
- Geographic Distribution
- Seller Distribution

Key Findings

Marketplace Growth

Order volume increased significantly throughout 2017 and remained relatively stable during 2018.

This trend indicates strong marketplace growth and increasing customer adoption during the analysis period.



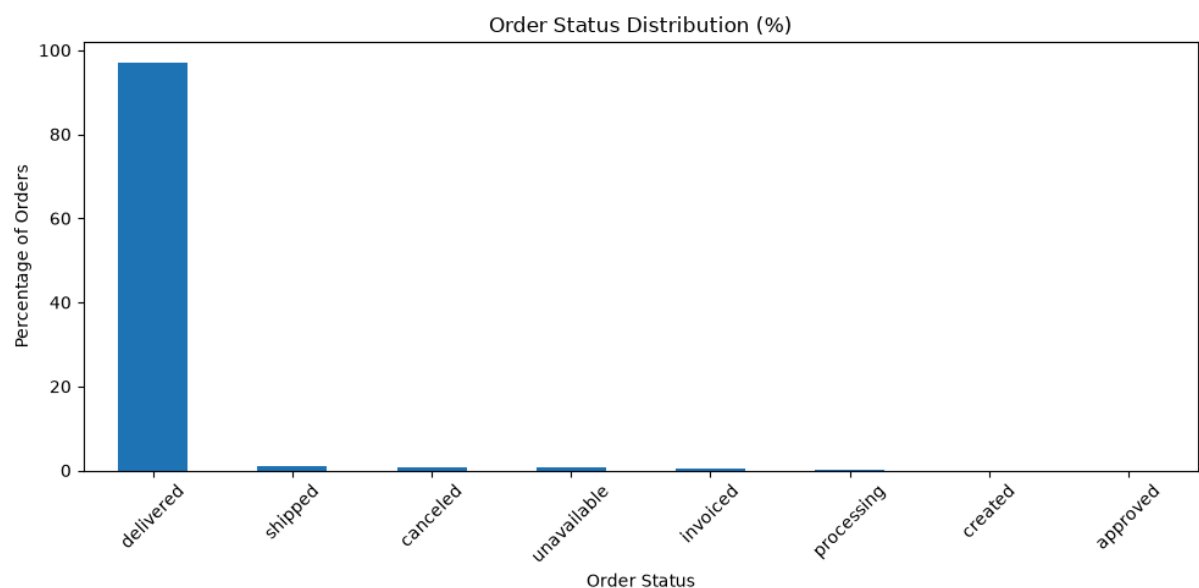
Order Fulfillment Performance

The marketplace demonstrated a high order completion rate.

Key Metrics

- 97.02% of orders were successfully delivered.
- Less than 3% of orders were associated with other statuses such as cancelled, unavailable, invoiced, or processing.

This suggests strong operational performance and fulfillment reliability.



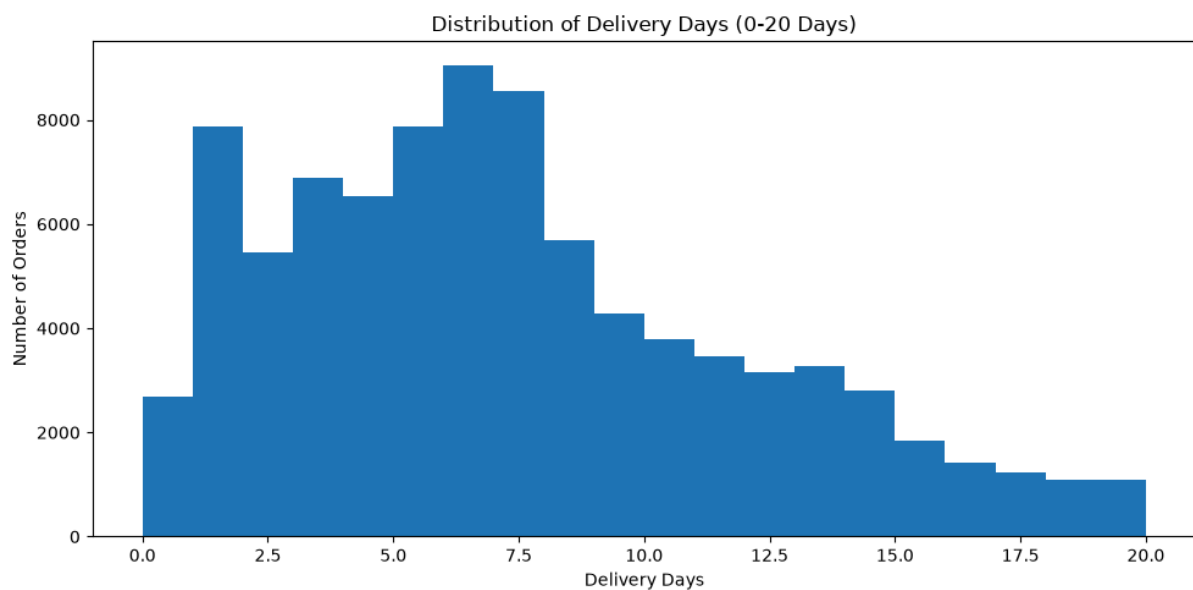
Delivery Performance

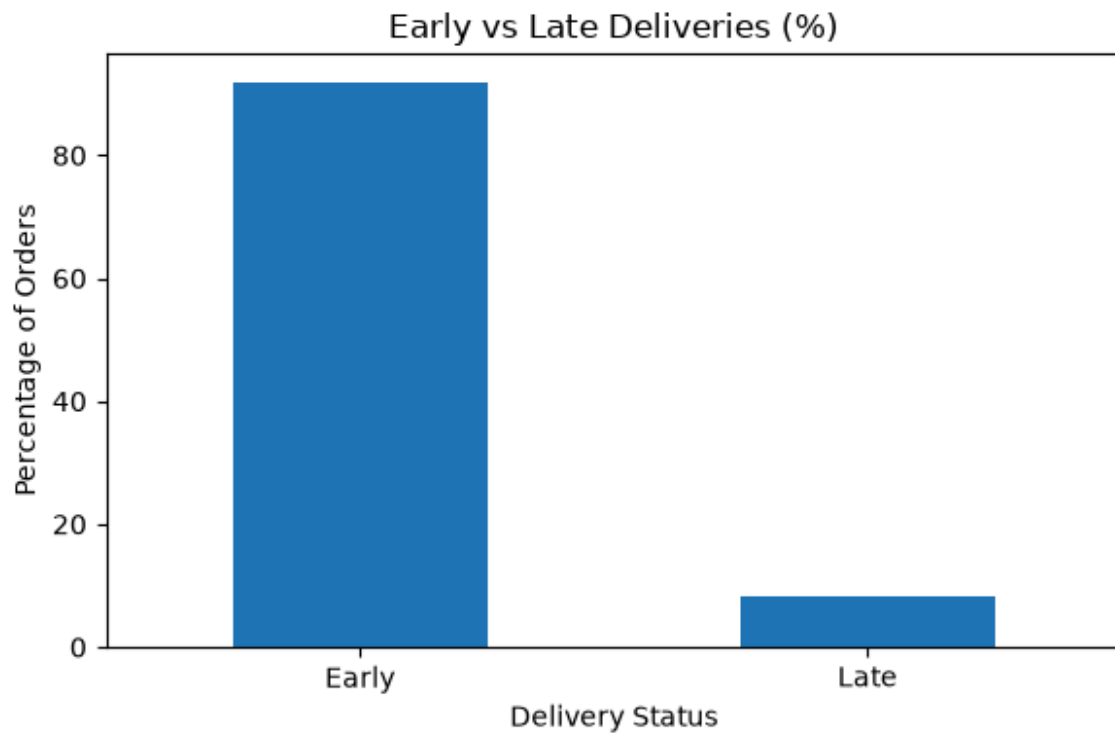
Delivery-related features created during feature engineering were used to evaluate logistics performance.

Key Metrics

- Average delivery time: 9.33 days
- Average fulfillment time: 12.56 days
- 91.89% of orders were delivered earlier than the estimated delivery date
- 8.11% of orders were delivered later than expected

These results indicate efficient logistics operations and conservative delivery estimates.





Customer Satisfaction

Customer reviews were analyzed to understand overall marketplace sentiment.

Key Metrics

- Average review score: 4.08
- 57.78% of reviews received 5-star ratings
- 77.07% of reviews were positive (4 or 5 stars)

Overall customer satisfaction was consistently high across the marketplace.



Revenue Performance

Revenue trends closely followed order volume trends.

Key observations included:

- Significant revenue growth throughout 2017
- Peak revenue observed during late 2017
- Stable revenue performance throughout much of 2018

The findings suggest that business growth was primarily driven by increasing transaction volume.

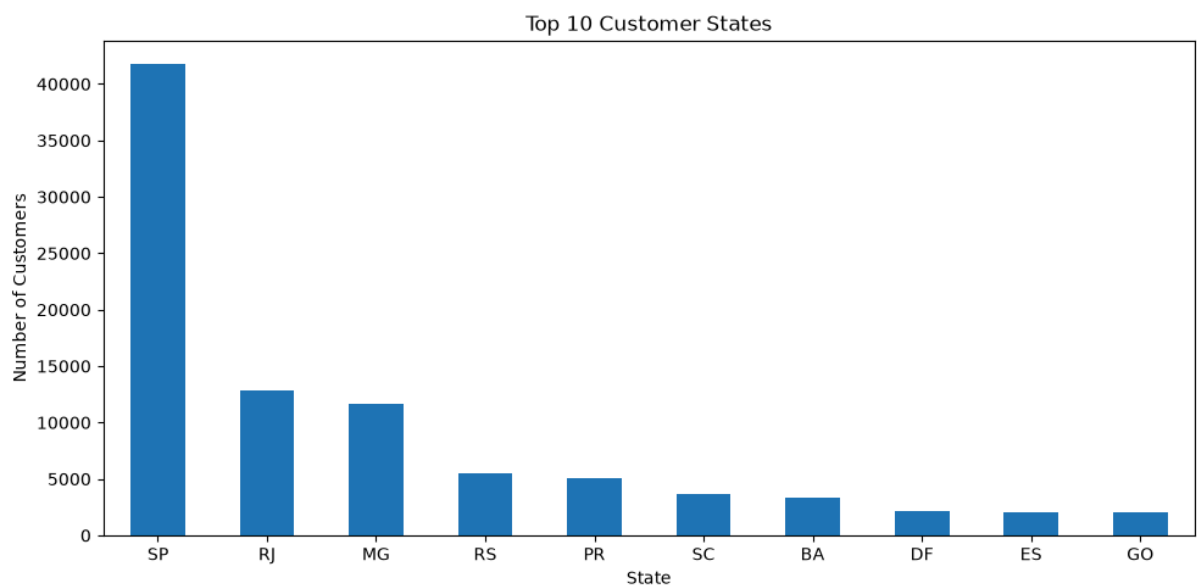
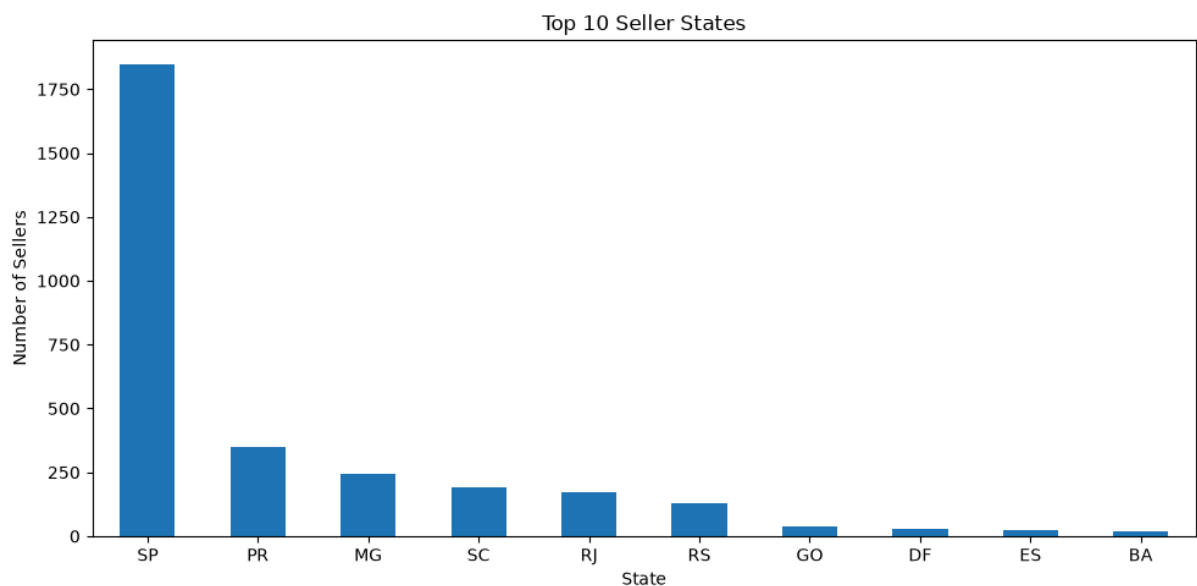
Geographic Distribution

Customer and seller activity were concentrated within a limited number of Brazilian states.

Key Observations

- São Paulo (SP) represented the largest customer market.
- São Paulo (SP) also contained the highest concentration of sellers.
- Customer demand and seller supply were heavily concentrated within Brazil's Southeast region.

This highlights the strategic importance of the Southeast region within the marketplace ecosystem.



Seller Distribution

Seller distribution was analyzed to understand the supply side of the marketplace.

Key Observations

- São Paulo (SP) contained the largest concentration of sellers.
- Paraná (PR) and Minas Gerais (MG) were the next largest seller regions.
- Seller activity was concentrated within a limited number of states.

- Seller distribution closely mirrored customer distribution.

The results suggest that marketplace supply is concentrated in the same regions that generate the highest customer demand, indicating a strong alignment between buyers and sellers.

Business Insights

The Python EDA phase revealed several important marketplace characteristics:

- Strong marketplace growth and adoption.
- High order fulfillment success rates.
- Efficient delivery operations.
- Positive customer sentiment.
- Stable revenue growth.
- Geographic concentration of both demand and supply.

These findings provided an initial understanding of marketplace performance and helped identify potential areas for deeper investigation during the SQL analysis phase.

Outcome

The Python EDA phase successfully transformed cleaned data into actionable business understanding.

The insights generated during this phase established a foundation for:

- SQL Exploratory Data Analysis
- KPI Development
- Hypothesis Testing
- Business Insight Generation
- Power BI Dashboard Development

The project is now prepared to transition into advanced SQL-based analysis.